REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Motion with Constant Acceleration Practice Questions

Grade 9 Advanced | Module 3 Lesson 2 | Motion with Constant Acceleration

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $v_f = v_i + at$; $\Delta x = v_i t + \frac{1}{2}at^2$; $v_f^2 = v_i^2 + 2a\Delta x$; $\Delta x = \frac{1}{2}(v_i + v_f)t$. Use only for constant acceleration.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 63–70.

P01. For constant acceleration, the velocity–time graph is:

A. Always horizontal B. Always a parabola
C. A straight line D. A circle

P02. Which relation does not require elapsed time explicitly?

A. $v_f = v_i + at$ B. $\Delta x = v_i t + \frac{1}{2}at^2$
C. $\Delta x = \frac{1}{2}(v_i + v_f)t$ D. $v_f^2 = v_i^2 + 2a\Delta x$

P03. A curved position–time graph for constant nonzero acceleration is:

A. A parabola B. Always a straight line
C. Always horizontal D. A vertical line

P04. For constant acceleration, average velocity over an interval equals:

A. The sum of the two speeds always B. Half the sum of initial and final velocities
C. Final velocity alone D. Acceleration multiplied by distance

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P05. Before applying the standard motion equations, you should check that:

A. Mass is zero B. Displacement is positive
C. Acceleration is constant over the interval D. Initial velocity is zero

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P06. An object has initial velocity +4 m/s and constant acceleration +3 m/s² for 3 s. Find its final velocity.

A. 13 m/s B. 26 m/s
C. 6.5 m/s D. -13 m/s

P07. An object starts at +4 m/s and accelerates at +2 m/s² for 4 s. Calculate its displacement.

A. 64 m B. 32 m
C. 16 m D. -32 m

P08. An object moves with initial speed 4 m/s and constant forward acceleration 2 m/s² through 6 m. Calculate its final speed.

A. 12.65 m/s B. 3.162 m/s
C. 6.325 m/s D. -6.325 m/s

P09. A vehicle travels at 8 m/s and brakes at a constant acceleration −2.0 m/s². Find its stopping distance.

A. 32 m B. 8 m
C. -16 m D. 16 m

P10. A car slows uniformly from +12 m/s to rest in 4 s. Calculate its acceleration.

A. -3 m/s² B. -6 m/s²
C. -1.5 m/s² D. 3 m/s²

P11. State the meaning of each symbol in vf = vi + at.

P12. Why is a sign convention important in one-dimensional motion?

P13. Which equation is useful when final velocity is unknown but displacement and time are needed?

P14. Why may total distance differ from displacement when using motion equations?

P15. What is the velocity–time slope for uniform acceleration?

P16. An object has initial velocity +6 m/s and constant acceleration +3 m/s² for 4 s. Find its final velocity.

P17. An object has initial velocity +8 m/s and constant acceleration +3 m/s² for 5 s. Find its final velocity.

P18. An object starts at +6 m/s and accelerates at +2 m/s² for 5 s. Calculate its displacement.

P19. An object starts at +8 m/s and accelerates at +2 m/s² for 6 s. Calculate its displacement.

P20. An object moves with initial speed 6 m/s and constant forward acceleration 2 m/s² through 9 m. Calculate its final speed.

P21. An object moves with initial speed 8 m/s and constant forward acceleration 2 m/s² through 12 m. Calculate its final speed.

P22. A vehicle travels at 12 m/s and brakes at a constant acceleration −2.0 m/s². Find its stopping distance.

P23. A vehicle travels at 16 m/s and brakes at a constant acceleration −2.0 m/s². Find its stopping distance.

P24. A car slows uniformly from +18 m/s to rest in 6 s. Calculate its acceleration.

P25. A car slows uniformly from +24 m/s to rest in 8 s. Calculate its acceleration.

P26. A car starts from rest with a = 2.0 m/s² for 5.0 s. Find final velocity and displacement.

P27. A ball moves at +12 m/s with acceleration −3.0 m/s². Find when it first stops and its displacement by then.

P28. Why can the formula average velocity = (vi + vf)/2 fail for nonuniform acceleration?

P29. Design a graph test for whether a cart has constant acceleration.

P30. A calculated stopping distance is negative when the car was moving positively and braking. What should be checked?